

Activity 4: ER Relationships

CSCI 340 / DATABASE SYSTEMS AND DATA MANAGEMENT

Interpret a scenario, justify the cardinality, draw the model

Name and collaborators

Each case gives you two entity sets and a relationship to model. Choose a relationship name, explain what its cardinality means, and draw an ER diagram. This activity is ungraded. Work individually or in a group, then compare your answers with someone else.

Use the lecture notation: rectangles for entity sets, a diamond for the relationship, arrows toward the “one” side, and double lines for required participation. Entity names are enough here; you do not need to choose keys.

Part A: Personal lockers at a gym

MEMBER is a gym member. LOCKER is a particular physical locker.

Model the gym’s current long-term locker rentals. Members may rent a personal locker for ongoing use. A member can hold one rental at a time, and that locker is reserved for that member alone. Unrented lockers remain in the gym’s inventory.

A1. Name the relationship and explain its cardinality

Read it in both directions: for one instance of each entity, how many instances of the other may be related? State the minimum and maximum, name the mapping cardinality, and justify your choices from the scenario. Make any additional assumption explicit.

A2. Draw your ER model

Include both entity sets, your named relationship, and the arrows and participation lines that express your answer above.

Part B: Home departments at a company

EMPLOYEE is a current employee. DEPARTMENT is an organizational unit within the company.

Model each employee's single home department on the company roster. A department brings together employees who do different jobs; a newly formed department may still be awaiting its first employee.

B1. Name the relationship and explain its cardinality

Read it in both directions: for one instance of each entity, how many instances of the other may be related? State the minimum and maximum, name the mapping cardinality, and justify your choices from the scenario. Make any additional assumption explicit.

B2. Draw your ER model

Include both entity sets, your named relationship, and the arrows and participation lines that express your answer above.

B3. Read the same relationship in reverse

How would you describe its mapping cardinality starting from DEPARTMENT instead of EMPLOYEE? Explain whether that changes the model.

Part C: Enrollment at a college

STUDENT is a registered student. SECTION is one scheduled offering of a course in the current term.

Model enrollment for the current term. Students choose sections to build their schedules, and each section has a class roster. The system includes students who have not enrolled yet and sections still awaiting their first students.

C1. Name the relationship and explain its cardinality

Read it in both directions: for one instance of each entity, how many instances of the other may be related? State the minimum and maximum, name the mapping cardinality, and justify your choices from the scenario. Make any additional assumption explicit.

C2. Draw your ER model

Include both entity sets, your named relationship, and the arrows and participation lines that express your answer above.

C3. Add a relationship attribute

The college records a `final_grade` for each student in each section they complete. Add this attribute to your diagram using a dashed line. Explain why storing one `final_grade` on STUDENT or one on SECTION would lose information. Give an example for each.

Compare and reflect

After all three cases, compare your models with another individual or group, including if you worked alone. Identify one difference or shared choice. Explain why you revised or kept your answer, referring to the scenario or an assumption you made.